WEEK 5 PROGRESS REPORT

Week 5 Progress Report

Medical Card Generator System

Organization: Fauji Cement Company Limited (FCCL) – HR Management Office

Project Title

Automated Medical Card Generator System

Introduction

During Week 5 of the internship at Fauji Cement Company Limited (FCCL), a business process within the HR Management Office was analyzed to identify repetitive and time-consuming tasks. After observing the existing workflow, it was found that the process of creating employee medical cards was completely manual. Based on this analysis, the development of an **Automated Medical Card Generator System** was initiated to improve efficiency, accuracy, and productivity.

Problem Statement

The HR Management Office at FCCL creates medical cards for employees and their family members. Previously, this process was performed manually using Adobe software. HR staff had to open the card template, edit every field individually, and copy information from medical forms into the card. Each card required manual adjustments, making the process slow, repetitive, and prone to human error.

Project Description

The Medical Card Generator System is a desktop-based application developed to automate the process of generating medical cards. Instead of manually editing each card, the system provides a pre-designed medical card template in which all fixed information is already available. Users simply enter the employee's or family member's details, and the software automatically generates a professional medical card within seconds.

The system also allows users to save and manage records efficiently and export the generated medical cards in multiple formats suitable for printing.

Goals

- Automate the medical card generation process.
- Reduce manual editing and repetitive work.
- Minimize human errors.
- Improve HR department productivity.
- Standardize the design of medical cards.
- Enable quick generation and printing of medical cards.

Objectives

- Analyze the existing manual workflow.
- Identify challenges faced by the HR department.
- Develop an easy-to-use desktop application.
- Store employee and family records securely.
- Generate medical cards automatically using predefined templates.
- Provide export options for printing and digital sharing.
- Improve the overall efficiency of the HR medical card management process.

Existing System

The existing workflow involved:

1. Receiving employee medical forms.
2. Opening Adobe software.
3. Editing the medical card manually.
4. Copying employee information field by field.
5. Saving and printing the completed card.

Limitations

- Highly time-consuming.
- Repetitive manual work.
- Higher probability of typing mistakes.
- Inconsistent formatting.
- Difficult to manage large numbers of medical cards.

Proposed Solution

An Automated Medical Card Generator System was designed to eliminate these issues. The software contains predefined card templates where all permanent information remains fixed. HR staff only enters employee or family details, and the application automatically generates the medical card with proper formatting.

The generated medical card can be saved, printed, and exported in different formats without requiring any manual editing.

Tools and Technologies Used

Programming Language

- Python

Frontend Technologies

- PySide6 (Qt for Python)

- Qt Widgets Framework
- Qt Designer (for designing user interfaces)

Backend Technologies

- Python
- MVC (Model–View–Controller) Architecture

MVC Components

- **Model:** Handles employee records, business logic, and database operations.
- **View:** Graphical User Interface (GUI) developed using Qt Widgets.
- **Controller:** Connects the user interface with business logic and processes user actions.

Database

- SQLite
- SQLite3 Library

Output & Export Features

- Generate professional medical cards.
- Export cards as **PNG (High Quality Image)**.
- Export cards as **PDF**.
- Support **DPI-based printing** for high-resolution print quality.

System Features

- User-friendly desktop interface.
- Pre-designed medical card templates.
- Employee information management.
- Family member information management.
- Automatic medical card generation.
- SQLite database integration.
- Fast searching and updating of records.
- Export to PNG format.
- Export to PDF format.
- High-resolution DPI-based printing support.
- Consistent and professional card layout.

Work Completed During Week 5

During Week 5, the following tasks were completed:

- Studied the existing manual process used by the HR Management Office.
- Identified the major issues and inefficiencies.
- Defined the project scope and requirements.
- Designed the proposed automated solution.
- Selected the technology stack.
- Planned the MVC architecture.
- Designed the database structure using SQLite.
- Created the initial user interface using PySide6 and Qt Widgets.
- Started implementing the Medical Card Generator application.

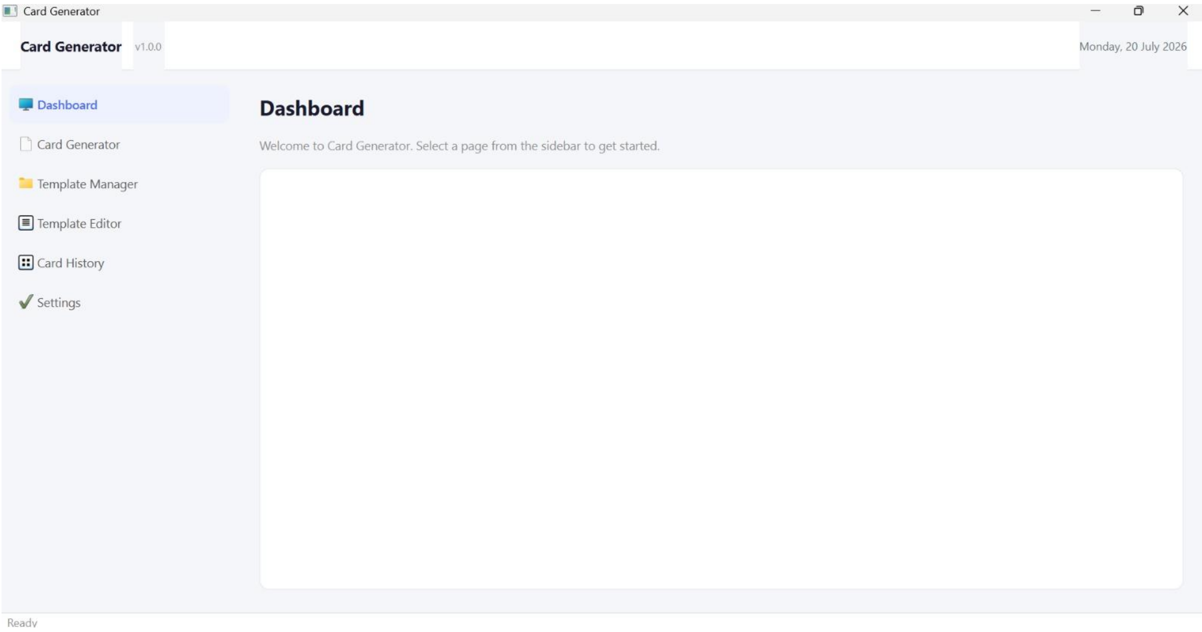
Expected Benefits

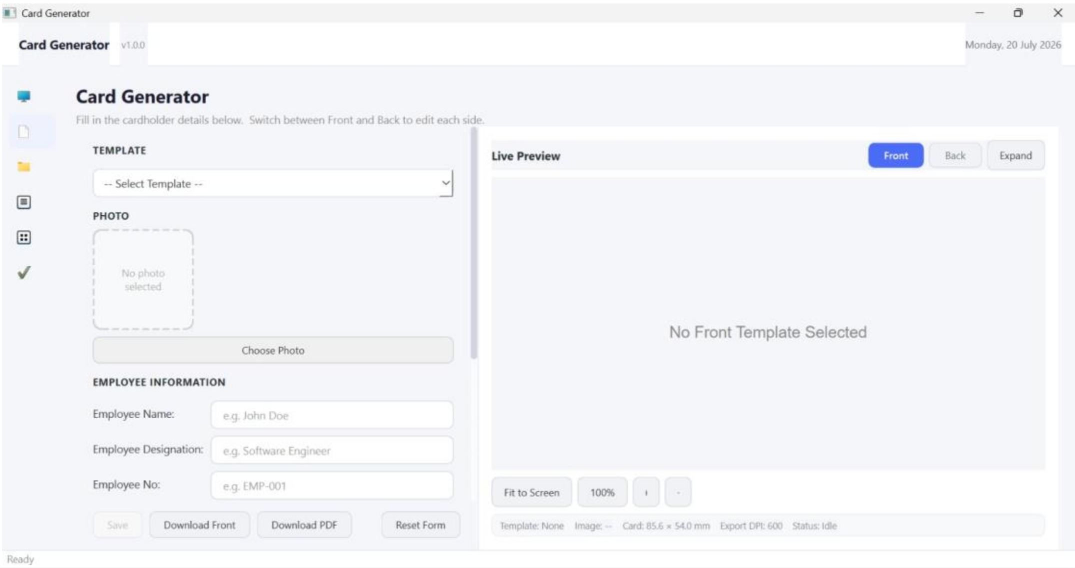
The proposed system will:

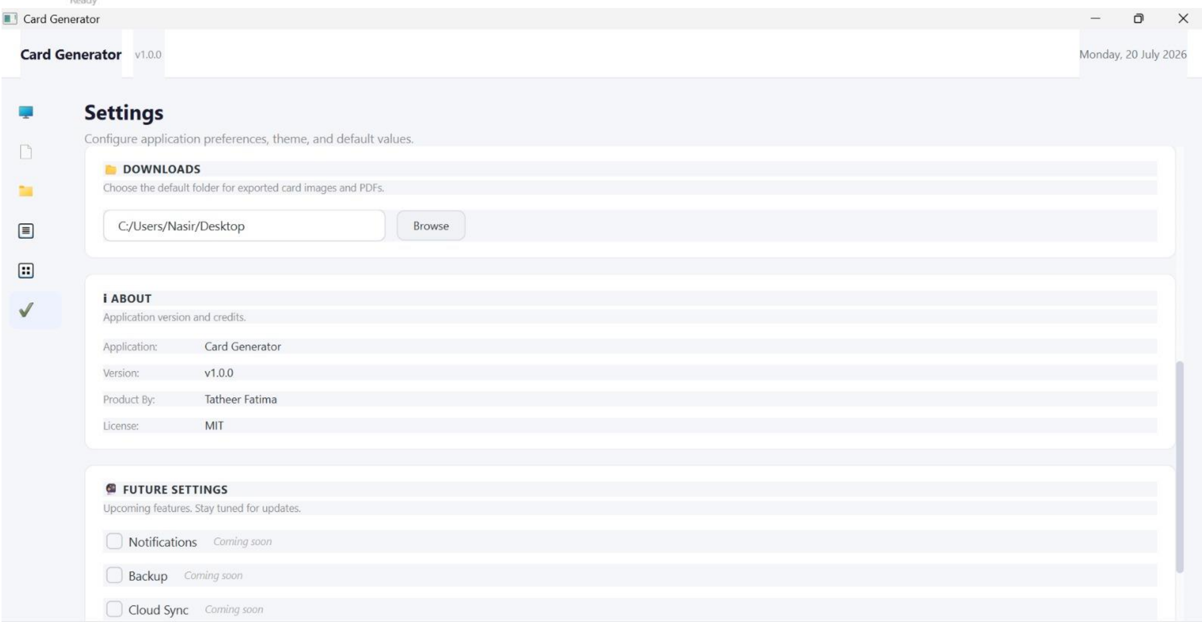
- Reduce medical card preparation time significantly.
- Eliminate repetitive manual editing.
- Improve data accuracy.
- Increase HR department efficiency.
- Produce standardized and professional medical cards.
- Simplify record management.
- Enable high-quality printing and digital sharing.

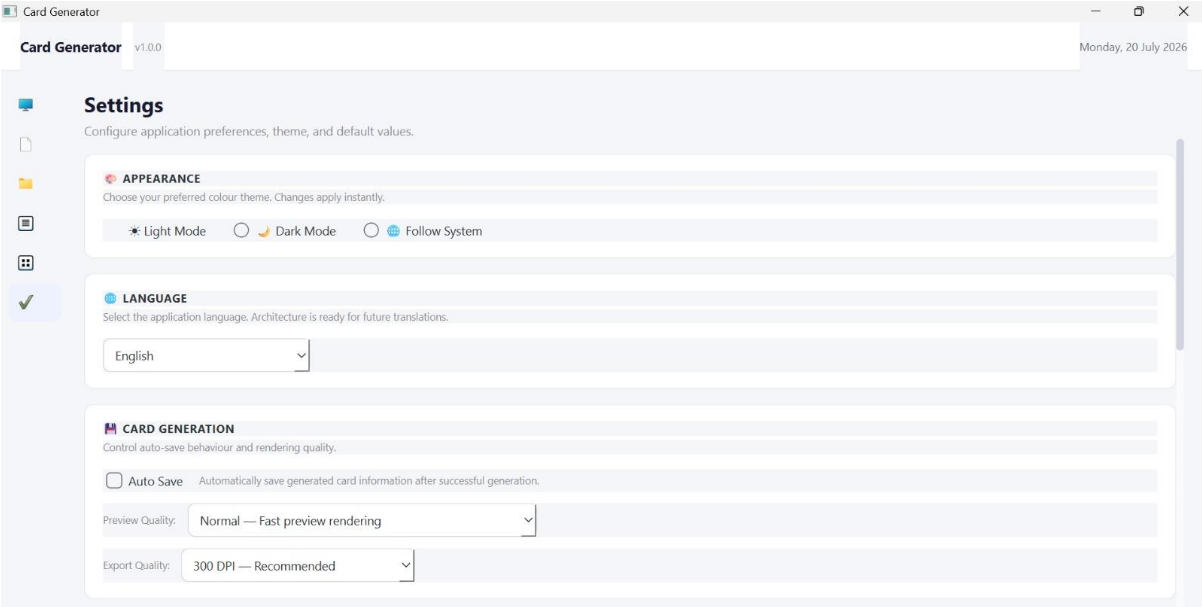
Conclusion

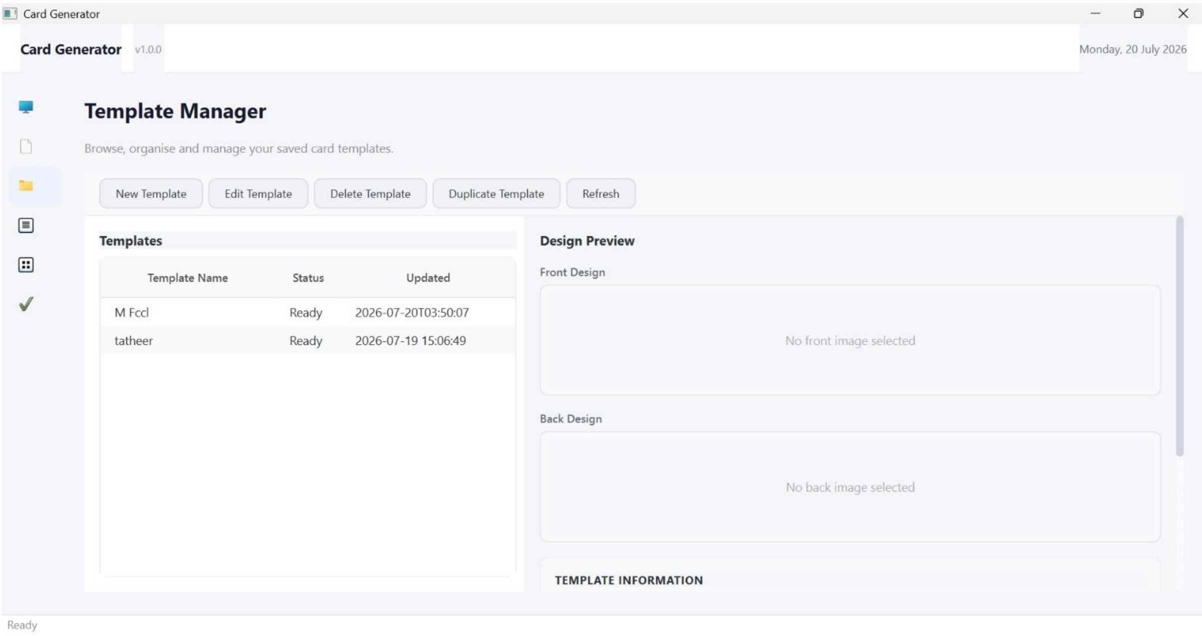
The Automated Medical Card Generator System is designed to modernize the medical card generation process within the HR Management Office at Fauji Cement Company Limited (FCCL). By replacing manual Adobe-based editing with an automated desktop application developed in Python using the MVC architecture, PySide6, Qt Widgets, and SQLite, the organization can save time, reduce errors, improve productivity, and generate professional-quality medical cards with support for PNG, PDF, and high-resolution DPI-based printing..

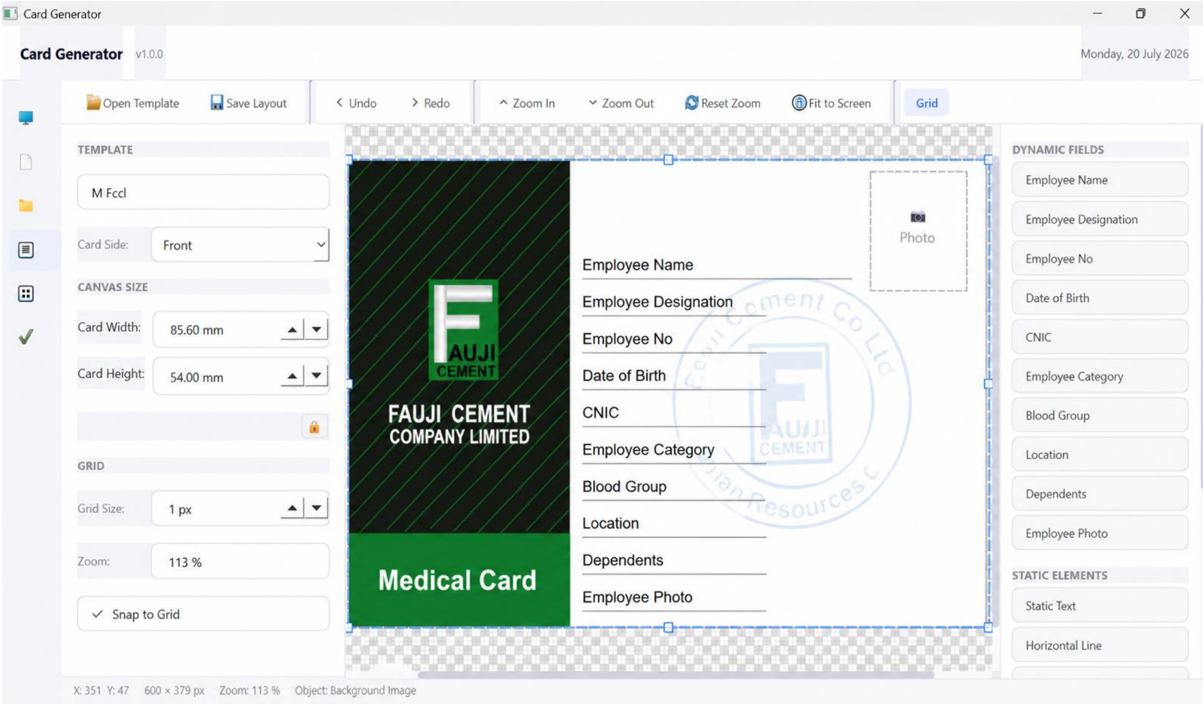

Card Generator
Card Generator v1.0.0
Monday, 20 July 2026
Dashboard
Card Generator
Template Manager
Template Editor
Card History
Settings
Dashboard
Welcome to Card Generator. Select a page from the sidebar to get started.
Ready

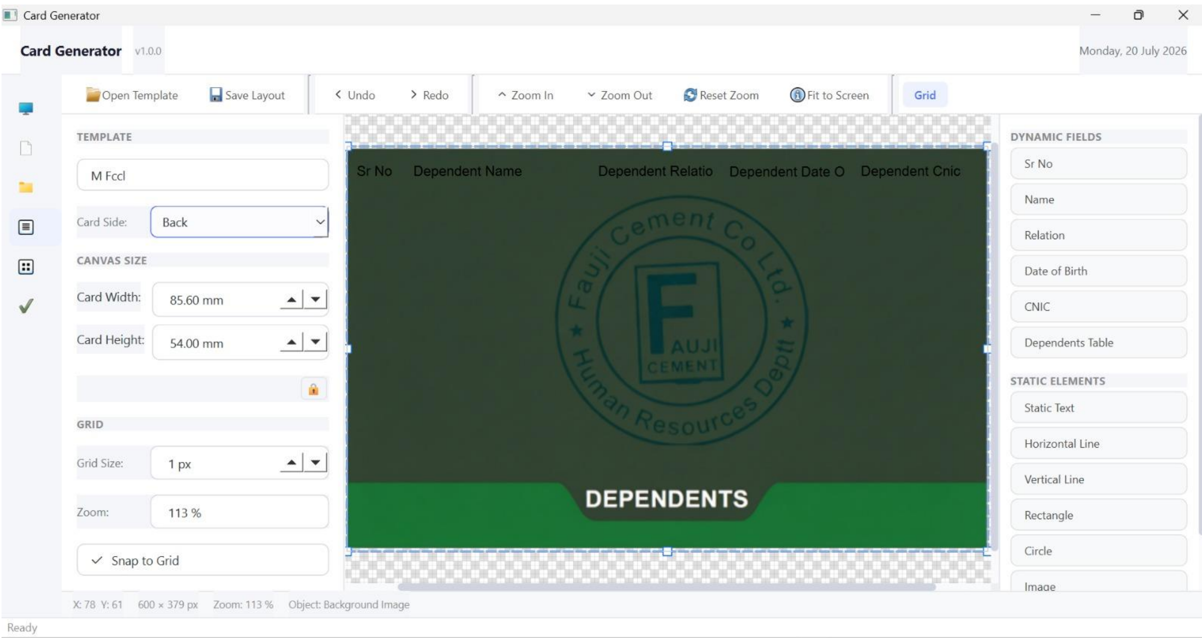

Card Generator
Card Generator v1.0.0
Monday, 20 July 2026
Card Generator
Fill in the cardholder details below. Switch between Front and Back to edit each side.
TEMPLATE
-- Select Template --
PHOTO
No photo selected
Choose Photo
EMPLOYEE INFORMATION
Employee Name:
e.g. John Doe
Employee Designation:
e.g. Software Engineer
Employee No:
e.g. EMP-001
Save
Download Front
Download PDF
Reset Form
Live Preview
Front
Back
Expand
No Front Template Selected
Fit to Screen
100%
Template: None Image: -- Card: 85.6 × 54.0 mm Export DPI: 600 Status: Idle
Ready


Card Generator
Card Generator v1.0.0
Monday, 20 July 2026
Settings
Configure application preferences, theme, and default values.
DOWNLOADS
Choose the default folder for exported card images and PDFs.
C:/Users/Nasir/Desktop
Browse
ABOUT
Application version and credits.
Application: Card Generator
Version: v1.0.0
Product By: Tatheer Fatima
License: MIT
FUTURE SETTINGS
Upcoming features. Stay tuned for updates.
Notifications Coming soon
Backup Coming soon
Cloud Sync Coming soon


Card Generator
Card Generator
v1.0.0
Monday, 20 July 2026
Settings
Configure application preferences, theme, and default values.
APPEARANCE
Choose your preferred colour theme. Changes apply instantly.
Light Mode
Dark Mode
Follow System
LANGUAGE
Select the application language. Architecture is ready for future translations.
English
CARD GENERATION
Control auto-save behaviour and rendering quality.
Auto Save
Automatically save generated card information after successful generation.
Preview Quality:
Normal — Fast preview rendering
Export Quality:
300 DPI — Recommended


Card Generator
Card Generator
v1.0.0
Monday, 20 July 2026
Template Manager
Browse, organise and manage your saved card templates.
New Template
Edit Template
Delete Template
Duplicate Template
Refresh
Templates
Template Name
Status
Updated
M Fccl
Ready
2026-07-20T03:50:07
tatheer
Ready
2026-07-19 15:06:49
Design Preview
Front Design
No front image selected
Back Design
No back image selected
TEMPLATE INFORMATION
Ready


Card Generator
Card Generator v1.0.0
Monday, 20 July 2026
Open Template
Save Layout
Undo
Redo
Zoom In
Zoom Out
Reset Zoom
Fit to Screen
Grid
TEMPLATE
M Fccl
Card Side: Front
CANVAS SIZE
Card Width: 85.60 mm
Card Height: 54.00 mm
GRID
Grid Size: 1 px
Zoom: 113 %
Snap to Grid
FAUJI CEMENT
COMPANY LIMITED
Medical Card
Photo
Employee Name
Employee Designation
Employee No
Date of Birth
CNIC
Employee Category
Blood Group
Location
Dependents
Employee Photo
DYNAMIC FIELDS
Employee Name
Employee Designation
Employee No
Date of Birth
CNIC
Employee Category
Blood Group
Location
Dependents
Employee Photo
STATIC ELEMENTS
Static Text
Horizontal Line
X: 351 Y: 47 600 × 379 px Zoom: 113 % Object: Background Image


Card Generator
Card Generator v1.0.0
Monday, 20 July 2026
Open Template
Save Layout
Undo
Redo
Zoom In
Zoom Out
Reset Zoom
Fit to Screen
Grid
TEMPLATE
M Fccl
Card Side: Back
CANVAS SIZE
Card Width: 85.60 mm
Card Height: 54.00 mm
GRID
Grid Size: 1 px
Zoom: 113 %
Snap to Grid
Sr No
Dependent Name
Dependent Relatio
Dependent Date O
Dependent Cnic
DEPENDENTS
DYNAMIC FIELDS
Sr No
Name
Relation
Date of Birth
CNIC
Dependents Table
STATIC ELEMENTS
Static Text
Horizontal Line
Vertical Line
Rectangle
Circle
Image
X: 78 Y: 61 600 × 379 px Zoom: 113 % Object: Background Image
Ready